

CHRISTOPHER CRILLY PIENAAH

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EDUCATION

Northeastern University , Toronto, ON <i>Master of Science in Analytics GPA: 3.96/4.0 Concentration: Applied Machine Intelligence</i>	Expected May 2026
Sir Sandford Fleming College , Toronto, ON <i>Graduate Certificate in Project Management GPA: 3.88/4.0</i>	July 2024
University of Nottingham , UK <i>Master of Science in Entrepreneurship, Innovation & Management</i>	December 2015
University for Development Studies , Tamale, Ghana <i>Bachelor of Arts in Economics & Entrepreneurship Studies</i>	December 2013

WORK EXPERIENCE

ICON Leadership Institute , Toronto, ON <i>AI/ML Product Strategist & Data Scientist</i>	May 2025 – Dec 2025
- Built a custom CRM system from scratch that automated lead tracking and cut manual work by 80% - Optimized GPT-4 prompts for document summarization improved relevance 20% while cutting token costs 15% - Automated 450K daily records processing through Python + SQL pipelines, reducing manual analysis from 10 hours to 1 hour - Created Tableau dashboards that executives use, turned raw data into insights that drove strategic decisions - Implemented data quality checks, ensuring 99.9% accuracy across multiple sources (saved hours of cleanup work)	
Chrisline Prints & Advertising , Kumasi, Ghana <i>Founder & Technical Lead</i>	May 2014 – June 2023
- Built Python + SQL pipelines that centralized data for 50+ SME clients, improving report accuracy 30% - Created Excel VBA dashboards that increased campaign ROI 30% through better targeting and timing - Led 10+ process improvement workshops that got teams to actually adopt the new automation (rare in consulting)	

FLAGSHIP PROJECTS

SME Growth Co-Pilot Google Kaggle AI Agents Competition (Top 50 of 11,494) <i>Multi-agent business intelligence system that replaces \$10K/month consultants</i>
- Built 6 AI agents that work together to analyze competitors, forecast financials, and generate growth strategies - Processes complex business analysis in 60 seconds with 100% success rate and 1ms response time - Found \$135K revenue opportunities for test case by identifying competitor gaps and pricing inefficiencies
LuminaMed AI Clinical Intelligence Platform for Radiology <i>An AI system that generates radiology reports doctors trust (92% accuracy, 3% hallucination rate)</i>
- Built a multi-agent verification that cross-checks findings against 50 medical journals before finalizing reports - Designed dual portals: one for radiologists (technical view) and one for patients (7th-grade reading level) - Implemented audit trails and compliance features for healthcare regulatory requirements from day one
Daavi Real Estate Platform for African Diaspora Investment <i>A property verification system that addresses the 68% trust barrier in diaspora real estate investment</i>
Built verification pipeline: government registry checks → legal doc validation → physical inspection workflows

- Created a recommendation engine that matches buyers with properties based on budget, location preferences, and investment goals
- Achieved 85% onboarding completion rate through clear cultural context education and investment guidance

Self-Healing Data Pipeline Platform | Autonomous DevOps AI (Live on AWS)

A multi-agent system that detects and fixes pipeline issues in <30 seconds, eliminating 3 am alerts

- Built 3 AI agents (Detective, Fixer, Critic) that autonomously remediate data pipeline failures with 85-92% confidence
- Deployed on AWS (App Runner + RDS), achieving 95% reduction in resolution time (2-8 hours → <1 minute)
- Saves enterprise teams \$1.5M+ annually by automating schema drifts, null spikes, and row count anomalies

BANKING ANALYTICS PORTFOLIO

Customer Churn Prediction | Built a model that predicts which bank customers will leave

- Analyzed 10,000 customer records and built a Random Forest model, achieving 86.93% accuracy
- Found that German customers had 45% churn rate vs. 16% overall recommended region-specific retention programs
- Used K-Means clustering to segment customers into Low/Medium/High risk groups for targeted intervention

Bank Marketing Propensity Model | Predicted which customers would buy term deposits

- Built XGBoost model on 11,162 customer records, achieving AUC 0.93 (excellent discrimination)
- Created an automated pipeline that showed 3x improvement in conversion rates by targeting high-propensity customers
- Discovered call duration and previous campaign success were the strongest predictors (not demographics like age)

TECHNICAL SKILLS & CERTIFICATIONS

AI/ML: LangChain, LangGraph, GPT-4, Claude API, Gemini 2.0 Flash, RAG, Vector DBs, Multi-Agent Systems, Prompt Engineering

Machine Learning: scikit-learn, XGBoost, Random Forest, Logistic Regression, K-Means, PCA, Hyperparameter Tuning

Full-Stack: FastAPI, Next.js, React, NestJS, Node.js, REST APIs, JWT/RBAC, WebSockets, Stripe, Slack API

Data: SQL (MySQL, PostgreSQL, SQL Server), Python (pandas, NumPy), ETL Pipelines, Star Schema, Data Warehousing

Cloud & Tools: AWS, Azure, Vercel, Docker, Git/GitHub, Tableau, Power BI, Excel (Advanced + VBA)

Certifications: 5-Day AI Agents Intensive (Kaggle + Google, Dec 2025) | AWS Generative AI (Udacity, 2025) | Applied AI & No-Code Development (DTTP, 2025) | Google Data Analytics (2024)

Bottom line: I build AI systems that move from idea → architecture → product → impact, especially in high-stakes domains where trust matters. Available June 2026 for full-time roles in AI/ML Engineering, Product Strategy, or Data Science.